

Tropical Cyclones

Large Scale Conditions for Formation

A. Aiyyer

MEAS, NCSU

Key References

- Emanuel, K., 2018: 100 Years of Progress in Tropical Cyclone Research.

Where do hurricanes form?

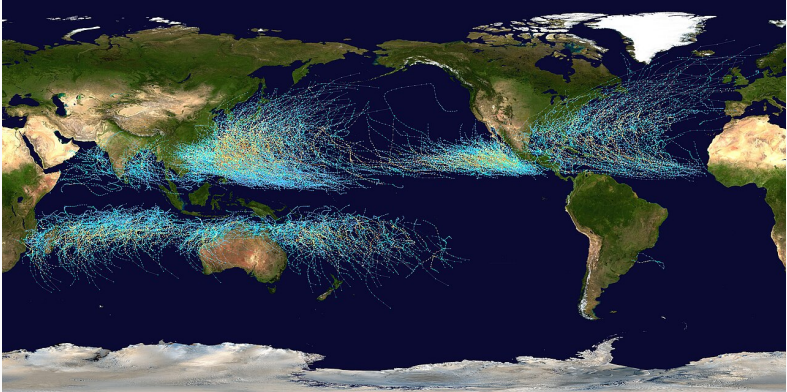


Figure 1: Source: wikipedia

Some key Questions

- Why do tropical cyclones form in specific regions of the world?
- Why is there a hurricane season in each basin? That is, why do the storms form in specific months of the year?
- What sets the number of storms per year?

- First We will examine some stats from CSU's webpage:
CSU Tropical Cyclones

Large scale conditions necessary for cyclogenesis

Over the past several decades, hundreds of research articles have been written on this issue.

The work by Bill Gray (CSU) in the 1960s was highly influential in shaping the general "consensus" regarding the state of the environment that is conducive for tropical cyclone formation.

We will now review the factors highlighted by Gray (1968) and subsequent early studies, and then throughout the course, we will see how this "consensus" has evolved with new research and data.

Large Scale factors

- Sea Surface Temperature exceeding 26°C
- Divergent flow in the upper troposphere,
- Moist lower to middle troposphere
- A Pre-existing cyclonic circulation (e.g., Easterly Wave)
- Low Vertical Wind shear

References: [\(click to be taken to the paper\)](#)

Gray 1968, Davis and Bosart 2003,

Sea Surface Temperature

See 2 "recent" papers:

Tory and Dare, 2015

McTaggart-Cowan et al. 2015

Sea Surface Temperature

- Roughly, a climatological value of 26 C appears to reasonably encapsulate the areas of most frequent tropical cyclone formation
- However, there are some basin-to-basin differences, with some storms forming at colder SST (e.g., the tropical transition type storms in the North Atlantic).
- There is a smooth increase in the number of tropical cyclones as SST increases. That is, there is no abrupt increase in the likelihood of storm formation above a fixed threshold.
- Tropical cyclone formation, even given SST higher than 26 C is still a rare phenomenon.

Wind Shear

Wind Shear: Typically defined as the magnitude of the vector difference of winds at upper and lower levels.

$$S = [(U_2 - U_1)^2 + (V_2 - V_1)^2]^{\frac{1}{2}} \quad (1)$$

Here subscript 2 refers to the upper-level (typically 200 hPa) and the 1 refers to the lower-level (typically 850 hPa).

- It is Generally accepted that high vertical wind shear precludes tropical cyclogenesis
- Some studies have found that cyclogenesis is unlikely when the magnitude of the 850–200-hPa shear exceeds 15 m/s (DeMaria et al. 2001)
- One could conclude that low wind shear is a necessary condition for cyclogenesis (we will examine the mechanisms in a later module).

Interesting results reported by: Nolan and McGauley. They ask two questions

- Is “some wind shear” more or less favorable for TC formation than no wind shear at all?

Statistical analyses of TC genesis events and numerical simulations show that light wind shear (2.5-5 m/s) is indeed the most favorable condition for genesis. Near-zero wind shear is not unfavorable - just very rare.

- Is easterly shear more or less favorable than westerly shear?

Numerical simulations suggest that, all other factors being equal, easterly shear is less favorable than westerly shear.

Wind Shear: Physical Mechanism

OK, Wind Shear appears to be detrimental to tropical cyclogenesis. Why is it so?

We will examine the physical mechanisms in detail in a later module (Tropical Cyclone Genesis and external interactions). Here are some ways that shear impacts genesis:

1. **Ventilation effect:** The core of a developing tropical storm needs to remain moist (close to saturation) in order to prevent evaporatively driven downdrafts. Vertical shear brings dry air inside the core which can lead to evaporation of rain and formation of cool downdrafts that transport this "dry" air downward and further preventing subsequent convection
2. **Dynamical effect:** Separation of upper and lower level circulations due to the shear - leading to a tilted structure.

Tropical Cyclone formation has been also shown to be sensitive to mid-tropospheric (~ 600 hPa) moisture

- A deep saturated column of air is pre-requisite for tropical cyclone formation (e.g., Bister and Emanuel 1997; Nolan 2011).
- The mid-troposphere is generally sub-saturated. So, if convection does fire up in this environment, the falling precipitation will evaporate. Downdrafts will ensue that will bring dry air into the boundary layer and prevent the initiation of additional convection from forming.
 - We saw that vertical shear acts to disrupt the cyclone by ventilating the core with sub-saturated air from outside.

Pre-existing Disturbance

In our current climate, tropical cyclones typically form within a pre-existing cyclonic disturbance

Common precursors to tropical cyclones are: African easterly waves (Atlantic); the ITCZ (eastern Pacific); equatorial waves (western Pacific); Monsoon trough/depressions (Indian Ocean/bay of Bengal).

Figure 2: Genesis Location of Tropical cyclones (1988–2022). Filled circles mark the ones that originated from AEWs. The color of the circles corresponds to the Saffir-Simpson category shown in the figure legend

Figure 3: Genesis Location of Tropical cyclones (1988–2022). Filled circles mark the ones that originated from AEWs. The color of the circles corresponds to the Saffir-Simpson category shown in the figure legend

Is a precursor really needed for cyclogenesis?

Over the past decade or so, exciting new research has shown that tropical cyclones can form out of self-aggregation of convection in an environment that is initially devoid of any precursor cyclonic disturbance.

Additional resources. Two interesting talks by Prof. Allison Wing:

MIT on Chaos and Climate: Convective Aggregation Clouds and Climate

ExoClimes V - Day 2: Allison Wing

Putting all the environmental controls together

- Is there a way for us to combine the effects of all these environmental factors that control cyclogenesis?
- What would be the advantage of a combined "variable"?

Following material taken from Emanuel (2018)

- Gray (1975, 1979): Developed an empirical relationship (A genesis index) between the frequency of tropical cyclogenesis and climatological properties of the large-scale environment.
- Other genesis indices have been developed in recent years (Emanuel and Nolan 2004; Emanuel 2010; Tippett et al. 2011; Bruyère et al. 2012).

Examples of a genesis potential index: Kerty et al. 2012

Emanuel 2018

Genesis Potential Index by Emanuel and Nolan (2004):

$$GP = |10^5 \eta|^{3/2} \left(\frac{RH}{50} \right)^3 \left(\frac{V_{pot}}{70} \right)^3 (1 + 0.1 V_{shear})^{-2} \quad (2)$$

where η is the absolute vorticity at 850 hPa (in s^{-1}), RH is the relative humidity at 700 hPa, V_{pot} is the potential intensity in ms^{-1} , and V_{shear} is the magnitude of the vertical wind shear (in ms^{-1}) between 850 hPa and 200 hPa.

Figure 4: Tropical cyclone count and area-averaged genesis potential for (a) Atlantic and (b) the eastern Pacific.

Figure 5: Relationship between the annual tropical cyclone count and area-averaged genesis potential in the (a) Atlantic and (b) the eastern Pacific. The magenta shows the LOWESS curve while the orange line shows the prediction using Poisson regression. The orange shading denotes the 95% confidence interval for the latter.

Why are genesis potentials useful?

A search on google scholar

Why are genesis potentials useful?

Some applications:

- For seasonal forecasting of tropical cyclone activity
- For a theoretical understanding of the combine effect of parameters controlling cyclogenesis
- As a benchmark for numerical model performance relative to observations
- Deducing changes in tropical cyclone frequency in future climate simulations